Name

Class

R. No.

- Mathematics
- IIT JEE / CBSE

- Test 1 (July 15 , 2010)
- Sets

- Maximum Marks 50
- Time 35 minutes

Instructions

1. **The test consists of five sections.**
2. **Each section carries 10 Marks and roughly 7 Minutes are allotted for each section.**
3. **Only the answers need to be written in the space provided, rough work should be done on the back of the pages.**

Section I

Q1: Which of the following are sets?

i) The 10 best filmmakers in the world.
ii) The natural satellites of Saturn.
iii) All the guitarists on the planet.
iv) The collection of all the prime numbers.
v) The collection of questions in this paper.
vi) The set of people having given the IIT-JEE.
vii) The set of all the components required to build a house.
viii) The collection of the most dangerous animals in the world.

4 Marks

Q2: Write the following sets in the Set-Builder Form.

i) (5,10,15,20,….50)
ii) (2,4,8,16,32…..1024)
iii) (7,11,13,17,19……………)

6 Marks

Section II

Q3: Which of the following are examples of a null set.

i) Set of all prime numbers divisible by 35.

ii) $\{ x : x$ is a point on the number line satisfying $x^2 = 3 \}$

2 Marks

Q4. State whether each of the following set is finite or infinite:

i) The set of all lines which are parallel to the x-axis

ii) The set of letters in the English alphabet

iii) The set of numbers which are multiple of 5.

iv) The set of animals living on the earth

v) The set of circles passing through the origin (0,0).

vi) The set of months of a year

vii) The set of positive integers greater than 100.

viii) The set of prime numbers less than 99.

4 Marks

Q5: In the following , state whether A = B or not. (i.e. the two sets are equal)

i) A = {-2,-3} and B = { x : x is a solution of $x^2 + 5x + 6 = 0$}

ii) A = { 1 ,2 ,3 } and B = {3, 1, 1, 1, 2}

4 Marks

Section III

Q6: Write down all the subsets of the following set

A = { 1 , {2,3}, {4,5,6}}

6 Marks

Q7 : What universal set would you choose for the following two sets

A = The set of all the right triangles in the x-y plane

B = The set of all the isosceles triangles in the x-y plane. 4 Marks

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Section IV

Q8: Find the union of these two sets (by writing down in the roster form)

A = { x : x is a natural number and 1 < x < 6 }

B = {x: x is an integer and 6<x<10} 4 Marks

Q9: If A = { x : x is a natural number} , B = { x : x is an even integer} and C = { x : x is a prime number }

Find $A \cap B \cap C$. 3 Marks

Q 10: If A = { 3, 6 ,9 , 12, 15, 18 , 21 } , B = {4 , 8 , 12 , 16 ,20 } , C = {2, 4 ,6 , 8 , 10 , 12 ,14 ,16} and D = {5 , 10 , 15 ,20}

Find

i) A – B
ii) B – C
iii) D – C

3 Marks

Section V

Q 11: Draw appropriate venn diagram of each of the following

i) $(A \cup B)'$
ii) $A' \cap B'$
iii) $(A \cap B)'$
iv) $A' \cup B'$
v) $A' \cap B$

10 Marks